MICROCOMPUTERS
&
INTERFACING

CHAPTER-1
HISTORY OF MICROPROCESSORS

Outline

- Introduction
- 4-Bit Microprocessors
- 8-Bit Microprocessors
- 16-Bit Microprocessors
- 32-Bit Microprocessors
- 64-Bit Microprocessors

Introduction

The Mechanical Age:

- The idea of a computing sys is not new,
- The idea of calculating with machine started before 500B.C, when the Babylonians invented Abacus, uses string of beads to perf. calc
- Not improved until 1642, when Blaise. Pascal invented a calculator constructed of gears and wheels…
- PASCAL programming language
- In 1823, Charles Babbage, commissioned by the Royal Astronomical Society of Great Britain to produce programmable calculating machine … Analytical Engine

Cont'd...

The Electrical Age:

- 1800s saw advent of Electric Motor(conceived by M. Faraday)
- a motor driven adding machines were invented based on mechanical calculator developed by B. Pascal
- Electrically driven mechanical calculators were common pieces of office equipment until 1970s, when small handheld electronic calculator introduced by Bomar, appeared
- Monroe was also a leading pioneer of electornic calculators

Cont'd…

- The 1st elec. Calculating machine, Z_3 was devp by Konrad Zuse in 1941, is a comp. sys
  - Used in Aircraft & Missile design (WWII)

- The 1st elec. Comp. sys was devp. to break secrete German military code by Alan Turing in 1943….Colossus….not programmable

- In 1946, the 1st GPPECS was devp. at Pennsylvania University.,,, called ENIAC
- Huge, 17,000 VTs, over 500 miles of wires, 30 tons, executes 100,000 IPs, programed by rewiring its ckts, needs frequent maintenance

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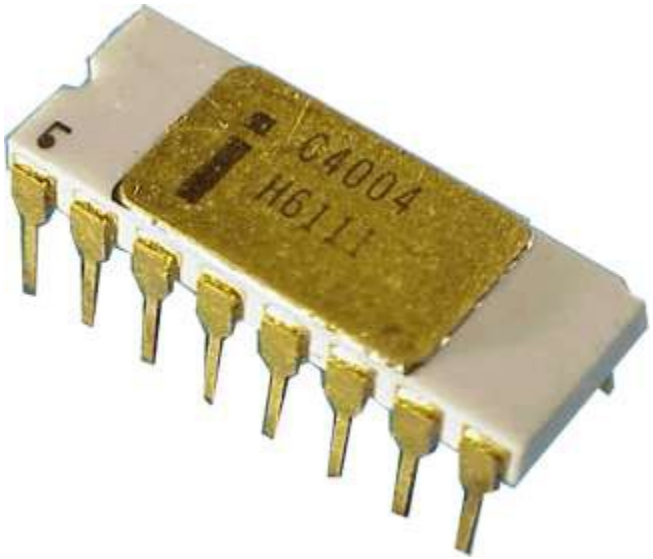
- Fairchild Semiconductors (founded in 1957) invented the first IC in 1959.
- In 1968, Robert Noyce, Gordan Moore, Andrew Grove resigned from Fairchild Semiconductors

- They founded their own company Intel (Integrated Electronics)
- Intel grown from 3 men start-up in 1968 to industrial giant by 1981
  - It had 20,000 employees and $188 million revenue

4-BIT MICROPROCESSORS

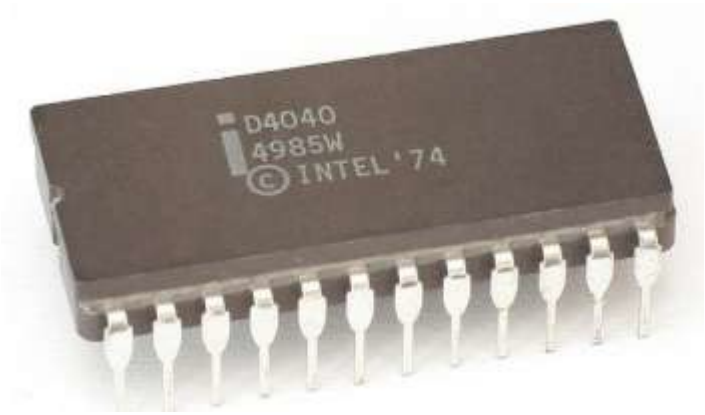
INTEL 4004:

- Introduced in 1971
- It was the first µP by Intel
- It was a 4-bit µP
- Its clock speed was 740KHz.
- It has 2,300 transistors
- It can execute around 60,000 instructions per second



4 data & 12 address bus, 45 instructions

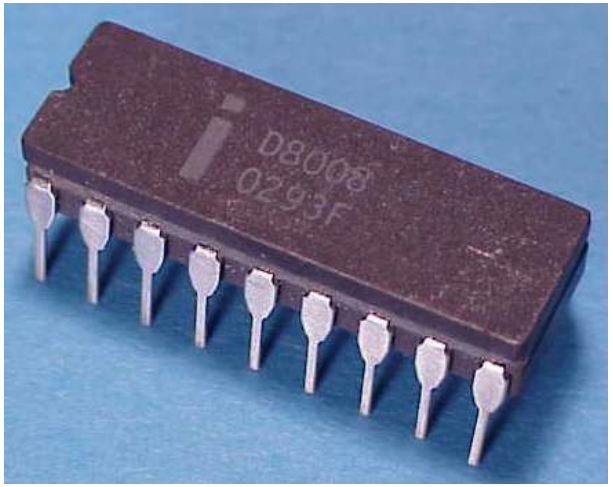
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INTEL 4040:



- Introduced in 1974
- It was also 4-bit µP
- Operates at higher speed than 4004
- Same size with 4004
- Has 3000 transistors
- Still alive in low-end applications like (microwave oven)

8-BIT MICROPROCESSORS

INTEL 8008



- Introduced in 1972
- It was the first 8-bit µP.
- Its clock speed was 500 KHz
- Could execute 50,000 instructions per second

8 data & 14 address bus,
Contains 3500 transistors

Cont'd…

INTEL 8080



8 data & 16 address bus

- Introduced in 1974
- The 1st modern 8-bit µP.
- Its clock speed was 2MHz
- It had 6,000 transistors
- Address 64K bytes of memory
- Was 10 times faster than 8008
- Execute 500,000 IPs

Cont'd…

Internal clock gen, sys controller, higher clock freq, still in use & will continue in the future

INTEL 8085

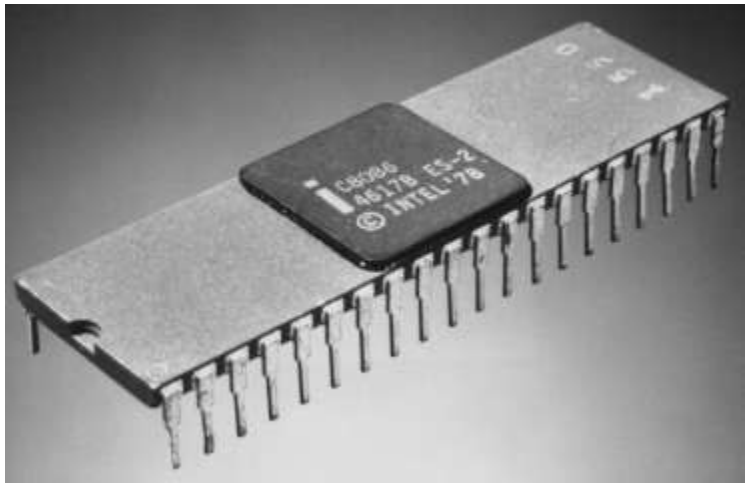


- Introduced in 1976
- It was also 8-bit µP

- Its clock speed is 3MHz
- Its data bus is 8-bit and address bus is 16-bit
- It had 6,500 transistors
- Execute 769,230 IPs
- It could access 64 KB of memory
- It had 246 instructions
- Over 100 million copies were sold

16-BIT MICROPROCESSORS

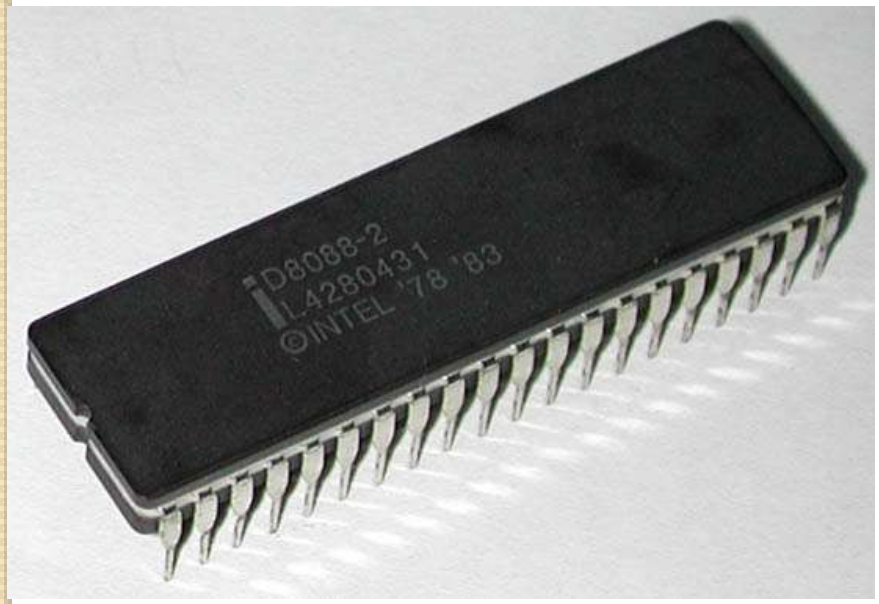
INTEL 8086



- Introduced in 1978
- It was first 16-bit μP
- Its clock speed is 4.77 MHz, 8 MHz and 10 MHz,
- Its data bus is 16-bit and address bus is 20-bit
  - Access 1 MB of memory
- It had 29,000 transistors
- Execute 2.5 million IPs
- It has 22,000 instructions
- Prefetches instructions
- It had ***Multiply*** and ***Divide*** instructions

Cont'd…

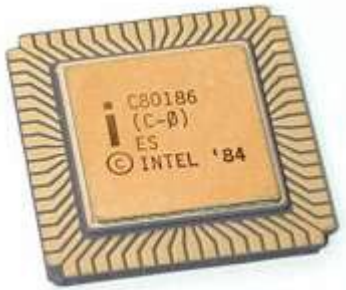
INTEL 8088

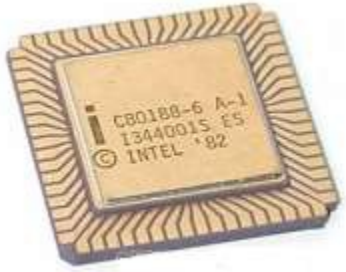


- Introduced in 1979.
- It is also 16-bit μP.
- Was created as a cheaper version of Intel's 8086
- It is a 16-bit processor with an 8-bit external bus
- Its clock speed is 510Mhz
- Execute 2.5 million IPs
- This chip became the most popular in the computer industry when IBM used it for its first PC

Cont'd…

INTEL 80186 & 80188





- Introduced in 1982
- They are 16-bit µPs

- Clock speed was 6 MHz
- 80188 was a cheaper version of 80186 with an 8-bit external data bus
- They have additional components like:
  - ✓ Interrupt Controller
  - ✓ Clock Generator
  - ✓ Local Bus Controller
  - ✓ Counters

Cont'd…

INTEL 80286



- Introduced in 1982
- It is 16-bit µP
- Its clock speed is 8MHz
- Its data bus is 16-bit and address bus is 24-bit
- Address 16MB of memory
- It has 134,000 transistors
- It can execute 4 MIPs
- Has additional instructions

32-BIT MICROPROCESSORS

INTEL 80386

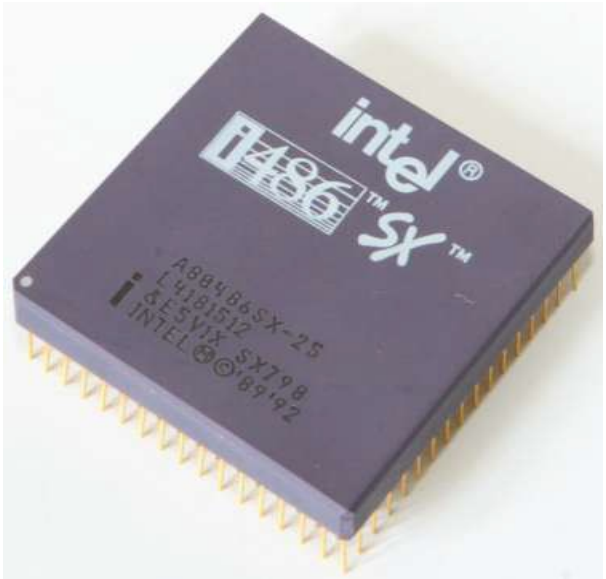


- Introduced in 1986
- It is the first 32-bit μP
- Its data bus is 32-bit and address bus is 32-bit
- Address 4 GB of memory
- It has 275,000 transistors
- Its clock speed varied from 16 MHz to 33 MHz depending upon the various versions
- Different versions:
  - ✓80386 EX
  - ✓80386 SX
  - ✓80386 SL
- Intel 80386 became the best selling μP in history

Cont'd…

50MIPs
50%
higher

INTEL 80486



- Introduced in 1989
- It is also 32-bit µP

- It has 1.2 million transistors
- Its clock speed varied from 16 MHz to 100 MHz depending upon the various versions
- It had five different versions:
  - 80486 DX
  - 80486 SX
  - 80486 DX2
  - 80486 SL
  - 80486 DX4
- 8-16 KB of cache memory was introduced

Cont'd...

INTEL PENTIUM



- Introduced in 1993
- It is also 32-bit μP

- It was originally named 80586
- Its clock speed was 66MHz
- Its data bus is 32-bit and address bus is 32-bit
- It can address 4 GB of memory
- Could execute 110 MIPs
- Cache memory:
  - 8 KB for instructions
  - 8 KB for data

Cont'd…

INTEL PENTIUM PRO



- Introduced in 1995
- It is also 32-bit µP
- It has 21 million transistors
- It is primarily used in server systems
- Cache memory:
  - ✓ 8 KB for instructions
  - ✓ 8 KB for data
- It has L2 cache of 256 KB

Cont'd...

INTEL PENTIUM II



- Introduced in 1997
- It is also 32-bit μP
- Its clock speed is 233 MHz to 500 MHz
- Can execute 333 million instructions per second
- MMX technology was supported
- L2 cache & processor were on one circuit

Cont'd…

INTEL PENTIUM II XEON



- Introduced in 1998
- It is also 32-bit μP
- It was designed for servers
- Its clock speed is 400MHz to 450 MHz
- L1 cache of 32 KB & L2 cache of 512KB, 1or 2 MB
- It could work with 4 Xeons in same system

Cont'd…

INTEL PENTIUM III



- Introduced in 1999
- It is also 32-bit µP
- Its clock speed varied

  from 500 MHz to 1.4GHz
- It has 9.5 million transistors

Cont'd...

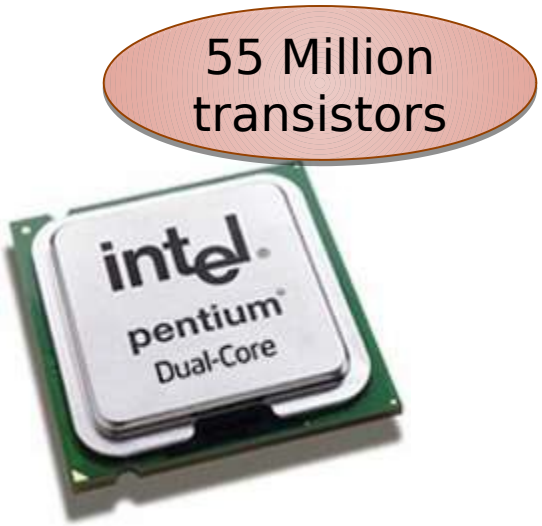
INTEL PENTIUM IV



- Introduced in 2000
- It is also 32-bit μP
- Its clock speed is from 1.3 GHz to 3.8 GHz
- L1 cache was of 32 KB & L2 cache of 256 KB
- It has 42 million transistors
- All internal connections were made from Al or Cu

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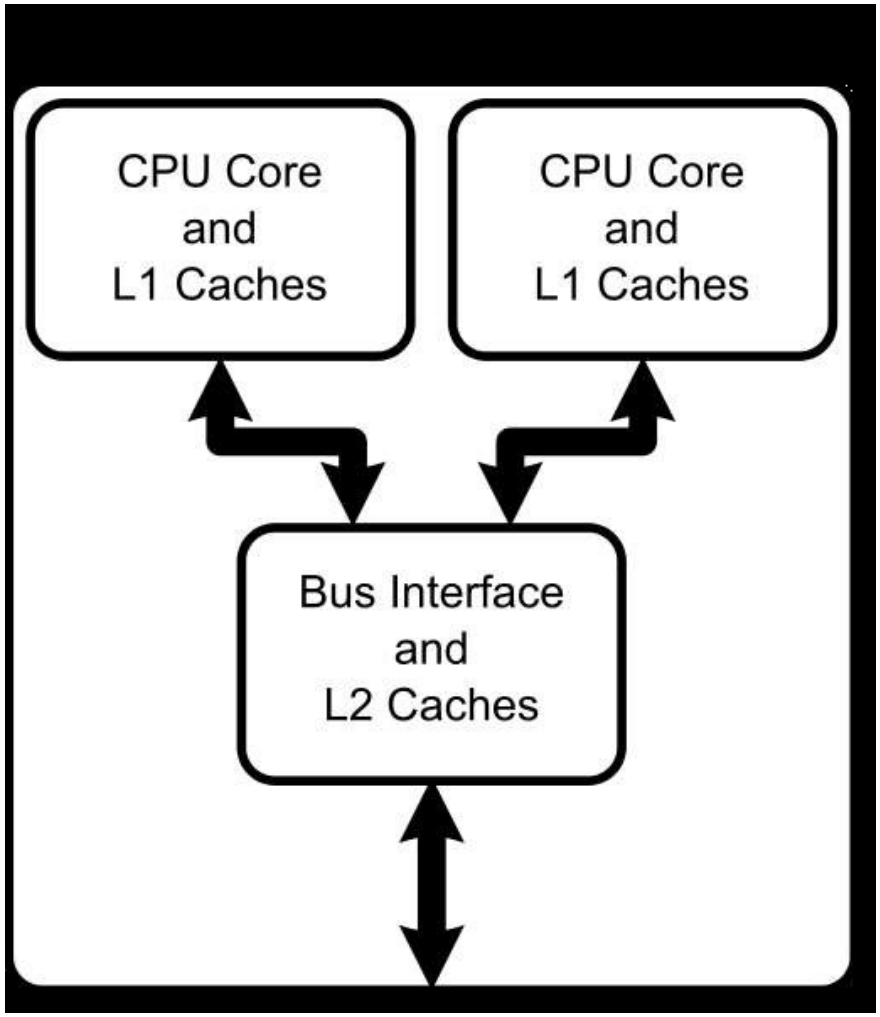
INTEL DUAL CORE



- Introduced in 2006
- It is 32-bit or 64-bit µP
- It has two cores

- Both the cores have their own internal bus and L1 cache, but share the external bus and L2 cache (*Next Slide*)
- It supports SMT tech.
- SMT: Simultaneously Multi-Threading
- E.g.: Adobe Photoshop supported SMT

Cont'd…


CPU Core
and
L1 Caches
CPU Core
and
L1 Caches
Bus Interface
and
L2 Caches

64-BIT MICROPROCESSORS

INTEL CORE 2



- Introduced in 2006.
- It is a 64-bit µP
- Its clock speed is from 1.2 GHz to 3 GHz
- It has 291 million transistors
- It has 64 KB of L1 cache per core and 4 MB of L2 cache
- It is launched in three different versions:
  - ✓Intel Core 2 Duo
  - ✓Intel Core 2 Quad
  - ✓Intel Core 2 Extreme

Cont'd...
INTEL CORE i3



- Introduced in 2010
- It is a 64-bit µP
- It has 2 physical cores
- Its clock speed is from 2.93GHz to 3.33 GHz
- It has 781 million transistors
- It has 64 KB of L1 cache per core, 512 KB of L2 cache and 4 MB of L3 cache

Cont'd…

INTEL CORE i5



- Introduced in 2009
- It is a 64-bit µP
- It has 4 physical cores
- Its clock speed is from 2.40 GHz to 3.60 GHz
- It has 781 million transistors
- It has 64 KB of L1 cache per core, 256 KB of L2 cache and 8 MB of L3 cache

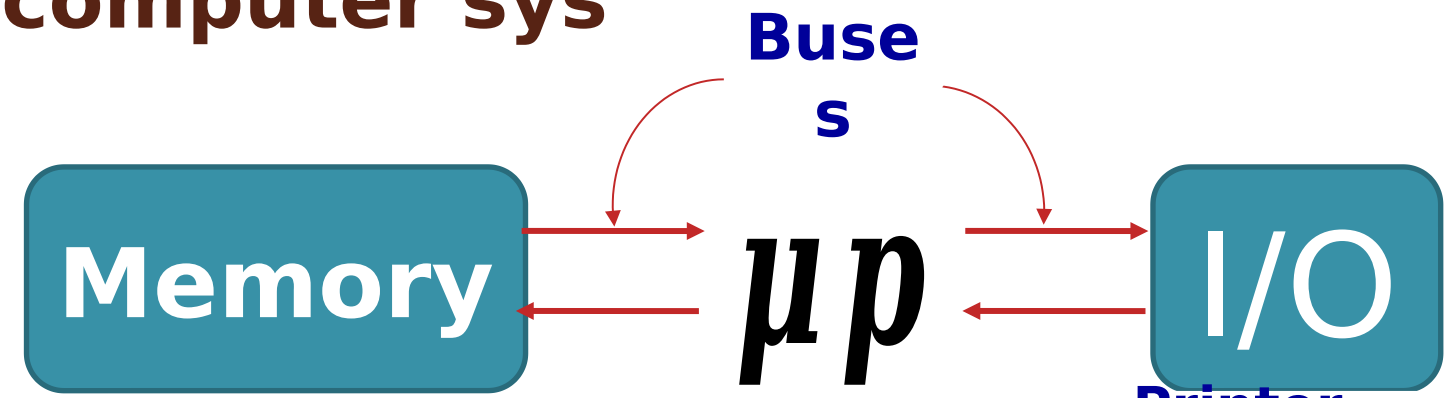
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INTEL CORE i7



- Introduced in 2008
- It is a 64-bit µP
- It has 4 physical cores
- Its clock speed is from 2.66GHz to 3.33 GHz
- It has 781 million transistors
- It has 64 KB of L1 cache per core, 256 KB of L2 cache and 8MB of L3 cache

Block diagram of based computer sys



Dynamic RAM
Static RAM
Cache
ROM
Flash Memory
EEPROM

8086
8088
80186
80188
80286
80386
80486
Pentium
Pentium Pro

Printer
Serial Comn.
Floppy disk drive
Hard disk drive
Mouse
CD-ROM drive
Plotter
Keyboard
Monitor
Tape backup
Scanner

QUESTIONS?